

Synthetic data generation - Phase 1

Features of the tabular synthetic data generation

Data Categories: Tabular, Textual

Supported data categories: Tabular.

Data clarity: High Fidelity, Noisy and Random

Domains: Healthcare, Audit, Finance, Retail

Sample Size supports 10,000 data records.

Feature settings:

Feature type: None, Numerical, Categorical, Mixed

Feature distribution: None, Normal, Poisson, Custom

Relationships & constraints:

Relationships: None, Normal, Poisson, Custom

Constraints: Define any constraints for the data generation

Output formats: CSV, JSON, Parquet

Schema:

Column Names

Datatypes ["Integer", "Float", "String", "Boolean", "Categorical", "Date", "Time", "DateTime"]

Field Description

Max Length (up to 5000 characters)

To support large data generation following strategies were implemented:

Retry mechanism, exponential backoff and sleep time.

- Used MAX_RETRIES=5 with a while loop to control retry attempts for each batch generation.
- Implemented robust response handling to prevent LLM truncation by using exponential backoff with a retry config of max_retries=5, initial_delay=1s, and backoff_factor=2, along with dynamic sleep time computation.

Dynamic Batch processing

The batch size estimation uses schema and prompt structure:

- A TokenEstimator class is used to estimate:
 - Schema token count
 - Prompt token count (with filled-in template)
 - Estimated output tokens per sample.
 - The estimator computes how many records can fit within the model's max_tokens limit.

Overall Flow of dynamic batch processing:

1. Estimate the schema token usage.
 2. Build the prompt and calculate its token count.
 3. Estimate output tokens per record.
 4. Use model input/output token caps to compute safe batch size.
- # Model Token Limits for GPT-4o-mini
 - MODEL_INPUT_LIMIT = 128000 # 128K tokens (input limit for GPT-4o-mini)
 - MODEL_OUTPUT_LIMIT = 16384 # 16,384 tokens (output limit for GPT-4o-mini)
5. Use this batch size in your generator's loop to dynamically split the data generation workload.

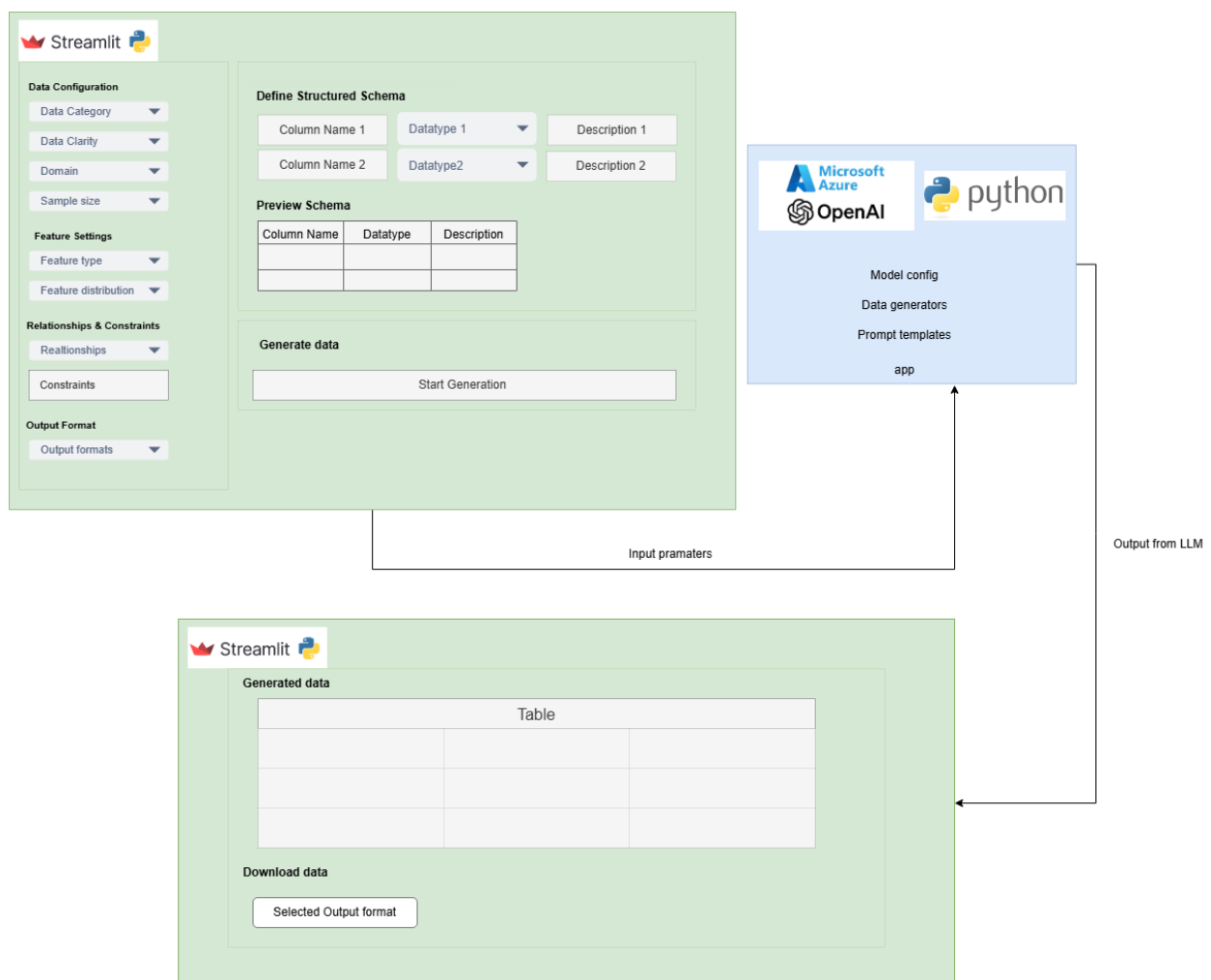


Fig 1: Synthetic data generation process flow

Testing various scenarios

Test 1: No of samples: 1000, No of columns: 9, No of records generated:1000

```
2025-05-07 13:17:37,240 - INFO - Initiating generation for category: Tabular
2025-05-07 13:17:37,240 - INFO - Using TabularDataGenerator.
2025-05-07 13:17:37,241 - INFO - Initializing Azure Open AI Model for tabular data
2025-05-07 13:17:37,241 - INFO - Initializing the Azure OpenAI model.
2025-05-07 13:17:37,242 - INFO - Azure OpenAI configuration loaded successfully.
2025-05-07 13:17:37,242 - INFO - No environment configuration found.
2025-05-07 13:17:37,243 - INFO - ManagedIdentityCredential will use IMDS
2025-05-07 13:17:37,243 - INFO - Token retrieved successfully.
2025-05-07 13:17:37,299 - INFO - Azure OpenAI LLM model initialized successfully.
2025-05-07 13:17:37,299 - INFO - Tokenizer initialized for model: gpt-4o-mini
2025-05-07 13:17:37,305 - INFO - Initializing the Azure OpenAI model.
2025-05-07 13:17:37,305 - INFO - Azure OpenAI configuration loaded successfully.
2025-05-07 13:17:37,305 - INFO - Tokenizer initialized for model: gpt-4o-mini
2025-05-07 13:17:37,307 - INFO - Schema token count: 309
2025-05-07 13:17:37,311 - INFO - Prompt token count: 2350
2025-05-07 13:17:37,315 - INFO - Prompt Sample token count: 2350
2025-05-07 13:17:37,315 - INFO - Using adaptive batch size: 50
2025-05-07 13:17:37,315 - INFO - Processing batch 1 with 50 records
2025-05-07 13:17:37,319 - INFO - Prompt token count: 2350
2025-05-07 13:17:37,329 - INFO - Request URL: 'http://169.254.169.254/metadata/identity/oauth2/token?api-version=REDACTED&resource=REDACTED'
Request method: 'GET'
```

```
2025-05-07 13:29:53,542 - INFO - Final DataFrame with 1000 records.
2025-05-07 13:29:53,542 - INFO - Total 1000 unique records generated.
2025-05-07 13:29:53,542 - INFO - Total 0 duplicate records skipped.
2025-05-07 13:29:53,542 - INFO - 0 batches failed.
2025-05-07 13:29:53,542 - INFO - Average records per batch: 50.00
2025-05-07 13:29:53,543 - INFO - Cumulative retry count: 0
2025-05-07 13:29:53,544 - INFO - Saving synthetic data as: tabular_data.csv
2025-05-07 13:29:53,566 - INFO - Synthetic tabular data saved to: D:\Users\vsr
_category_pipeline\Data\tabular_synthetic_data\tabular_20250507_132953.csv
2025-05-07 13:29:53,567 - INFO - Synthetic Tabular data saved successfully
2025-05-07 13:29:53,568 - INFO - Data generation completed successfully.
```

Start Generation

Data generated successfully! 1000 records generated. Here's a Preview:

	audit_id	audit_type	auditor_name	audit_date	audit_question	audit_answer
0	AUDIT-001	Financial Audit	Rajesh Kumar	2023-01-15	Are all financial statements in compliance with the latest accounting standards?	Yes, all financial statements were reviewed and f
1	AUDIT-002	Operational Audit	Anita Sharma	2023-01-20	Is the inventory management process efficient?	The process is efficient but needs minor improve
2	AUDIT-003	Compliance Audit	Suresh Patel	2023-02-05	Are all employees trained on compliance regulations?	Most employees are trained, but some need refrn
3	AUDIT-004	Quality Audit	Meena Gupta	2023-02-10	Is the product quality meeting customer expectations?	Quality is good, but feedback indicates room for
4	AUDIT-005	Environmental Audit	Vikram Singh	2023-02-15	Are waste disposal methods compliant with regulations?	Yes, all methods are compliant.
5	AUDIT-006	IT Audit	Priya Desai	2023-03-01	Are cybersecurity measures adequate?	Cybersecurity measures are in place but require
6	AUDIT-007	Health & Safety Audit	Amit Verma	2023-03-05	Are health and safety protocols being followed?	Protocols are mostly followed with minor lapses
7	AUDIT-008	Financial Audit	Nisha Rao	2023-03-10	Is there any discrepancy in the financial records?	Discrepancies were found and are being investig
8	AUDIT-009	Operational Audit	Rahul Choudhury	2023-03-15	Are operational procedures documented and followed?	Most procedures are documented, but some are
9	AUDIT-010	Quality Audit	Deepak Nair	2023-03-20	Is the defect rate within acceptable limits?	The defect rate is higher than acceptable limits.

2. No of samples:100, No of columns: 5, No of samples generated: 100

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2025-05-07 14:32:43,791 - INFO - Using TabularDataGenerator.
2025-05-07 14:32:43,791 - INFO - Initializing Azure Open AI Model for tabular data
2025-05-07 14:32:43,792 - INFO - Initializing the Azure OpenAI model.
2025-05-07 14:32:43,792 - INFO - Azure OpenAI configuration loaded successfully.
2025-05-07 14:32:43,793 - INFO - No environment configuration found.
2025-05-07 14:32:43,793 - INFO - ManagedIdentityCredential will use IMDS
2025-05-07 14:32:43,794 - INFO - Token retrieved successfully.
2025-05-07 14:32:43,926 - INFO - Azure OpenAI LLM model initialized successfully.
2025-05-07 14:32:43,926 - INFO - Tokenizer initialized for model: gpt-4o-mini
2025-05-07 14:32:43,957 - INFO - Initializing the Azure OpenAI model.
2025-05-07 14:32:43,968 - INFO - Azure OpenAI configuration loaded successfully.
2025-05-07 14:32:43,970 - INFO - Tokenizer initialized for model: gpt-4o-mini
2025-05-07 14:32:43,972 - INFO - Schema token count: 151
2025-05-07 14:32:43,978 - INFO - Prompt token count: 1488
2025-05-07 14:32:43,981 - INFO - Prompt Sample token count: 1488
2025-05-07 14:32:43,981 - INFO - Using adaptive batch size: 81
2025-05-07 14:32:43,982 - INFO - Processing batch 1 with 81 records

```

```

?api-version=2024-05-01-preview "HTTP/1.1 200 OK"
2025-05-07 14:33:26,268 - INFO - LLM Response token count: 5205
2025-05-07 14:33:26,270 - INFO - Processing batch 2 with 19 records
2025-05-07 14:33:26,273 - INFO - Prompt token count: 1488
2025-05-07 14:33:36,314 - INFO - HTTP Request: POST https://ai1lameuse2npdopenai.openai.azure.com/openai/deployment
?api-version=2024-05-01-preview "HTTP/1.1 200 OK"
2025-05-07 14:33:36,318 - INFO - LLM Response token count: 1398
2025-05-07 14:33:36,319 - INFO - Final DataFrame with 100 records.
2025-05-07 14:33:36,320 - INFO - Total 100 unique records generated.
2025-05-07 14:33:36,320 - INFO - Total 0 duplicate records skipped.
2025-05-07 14:33:36,320 - INFO - 0 batches failed.
2025-05-07 14:33:36,320 - INFO - Average records per batch: 50.00
2025-05-07 14:33:36,321 - INFO - Cumulative retry count: 0
2025-05-07 14:33:36,321 - INFO - Saving synthetic data as: tabular_data.csv
2025-05-07 14:33:36,327 - INFO - Synthetic tabular data saved to: D:\Users\vsreedharan\Documents\Test_AI_POC\deloitte_audit_category_pipeline\Data\tabular_synthetic_data\tabular_20250507_143336.csv

```

Data generated successfully! 100 records generated. Here's a Preview:

	audit_id	audit_type	audit_question	audit_answers
3	AUDIT-004	IT Audit	Are financial data systems secure from unauthorized access?	Yes, but there are some vulnerabilities identified in the current security protocols.
4	AUDIT-005	Forensic Audit	Have there been any irregularities in financial transactions?	Minor irregularities were found, but they were not material.
5	AUDIT-006	Tax Audit	Are all tax returns filed accurately and on time?	All tax returns were filed accurately, but one was late due to administrative error.
6	AUDIT-007	Financial Audit	How are financial risks assessed and managed?	Risks are assessed annually, but documentation could be improved.
7	AUDIT-008	Compliance Audit	Are there policies in place for financial reporting?	Yes, there are policies, but they are not consistently followed.
8	AUDIT-009	Operational Audit	Is there an efficient process for financial transactions?	The process is generally efficient, but there are bottlenecks in approval stages.
9	AUDIT-010	IT Audit	Are backups of financial data regularly performed?	Backups are performed, but the frequency needs to increase.
10	AUDIT-011	Forensic Audit	Is there evidence of financial mismanagement?	No substantial evidence of mismanagement was found.
11	AUDIT-012	Tax Audit	Are tax deductions claimed appropriately?	Most deductions are claimed properly, with a few discrepancies noted.

3. No of samples:2000, No of columns: 10, No of samples generated: 1802

```
2025-05-07 16:08:28,583 - INFO - Initiating generation for category: Tabular
2025-05-07 16:08:28,583 - INFO - Using TabularDataGenerator.
2025-05-07 16:08:28,583 - INFO - Initializing Azure Open AI Model for tabular data
2025-05-07 16:08:28,584 - INFO - Initializing the Azure OpenAI model.
2025-05-07 16:08:28,584 - INFO - Azure OpenAI configuration loaded successfully.
2025-05-07 16:08:28,585 - INFO - No environment configuration found.
2025-05-07 16:08:28,586 - INFO - ManagedIdentityCredential will use IMDS
2025-05-07 16:08:28,586 - INFO - Token retrieved successfully.
2025-05-07 16:08:28,724 - INFO - Azure OpenAI LLM model initialized successfully.
2025-05-07 16:08:28,725 - INFO - Tokenizer initialized for model: gpt-4o-mini
2025-05-07 16:08:28,733 - INFO - Initializing the Azure OpenAI model.
2025-05-07 16:08:28,733 - INFO - Azure OpenAI configuration loaded successfully.
2025-05-07 16:08:28,734 - INFO - Tokenizer initialized for model: gpt-4o-mini
2025-05-07 16:08:28,735 - INFO - Schema token count: 298
2025-05-07 16:08:28,740 - INFO - Prompt token count: 2293
2025-05-07 16:08:28,747 - INFO - Prompt Sample token count: 2293
2025-05-07 16:08:28,747 - INFO - Using adaptive batch size: 52
2025-05-07 16:08:28,747 - INFO - Processing batch 1 with 52 records
2025-05-07 16:08:28,751 - INFO - Prompt token count: 2293
2025-05-07 16:08:28,757 - INFO - Request URL: 'http://169.254.169.254/metadata/identity/oauth2/token?api-version=REDACTED&resource=REDACTED'
```

```
api-version=2024-05-01-preview http://169.254.169.254/metadata/identity/oauth2/token?api-version=REDACTED&resource=REDACTED'
2025-05-07 16:31:25,213 - INFO - LLM Response token count: 564
2025-05-07 16:31:25,219 - INFO - Final DataFrame with 1802 records.
2025-05-07 16:31:25,219 - INFO - Total 1802 unique records generated.
2025-05-07 16:31:25,219 - INFO - Total 0 duplicate records skipped.
2025-05-07 16:31:25,220 - INFO - 0 batches failed.
2025-05-07 16:31:25,220 - INFO - Average records per batch: 36.78
2025-05-07 16:31:25,220 - INFO - Cumulative retry count: 0
2025-05-07 16:31:25,221 - INFO - Saving synthetic data as: tabular_data.csv
2025-05-07 16:31:25,246 - INFO - Synthetic tabular data saved to: D:\Users\vsree
_category_pipeline\Data\tabular_synthetic_data\tabular_20250507_163125.csv
2025-05-07 16:31:25,246 - INFO - Synthetic Tabular data saved successfully
2025-05-07 16:31:25,247 - INFO - Data generation completed successfully.
```

Data generated successfully! 1802 records generated. Here's a Preview:

	Patient name	Patient age	Department visited	Name of the doctor	Medical history	Diagnosis	Prescribed treatment
0	Alice Johnson	34	Cardiology	Dr. Emily Carter	Hypertension, family history of heart disease.	Essential Hypertension	Lisinopril 10mg daily, lifes
1	John Smith	45	Neurology	Dr. Michael Brown	Migraines, no significant past medical history.	Chronic Migraine	Topiramate 50mg daily.
2	Maria Garcia	29	Dermatology	Dr. Sarah Wilson	No significant history.	Acne vulgaris	Benzoyl peroxide gel, oral
3	David Lee	52	Endocrinology	Dr. Linda Taylor	Type 2 diabetes, obesity.	Uncontrolled Type 2 Diabetes	Metformin 1000mg twice c
4	Sophia Martinez	38	Gynecology	Dr. Jennifer Davis	Irregular menstrual cycles.	Polycystic Ovary Syndrome	Hormonal contraceptives.
5	James Wilson	60	Oncology	Dr. Robert Anderson	Smoker for 30 years, family history of lung cancer.	Lung Cancer	Chemotherapy and radiati
6	Linda Thompson	41	Orthopedics	Dr. Christopher White	Previous knee surgery.	Knee Osteoarthritis	Physical therapy and NSAID
7	Michael Brown	50	Gastroenterology	Dr. Patricia Harris	History of GERD.	Gastroesophageal Reflux Disease	Omeprazole 20mg daily.
8	Patricia Miller	27	Pediatrics	Dr. Nancy Clark	Asthma since childhood.	Asthma	Albuterol inhaler as needed
9	Robert Davis	33	Psychiatry	Dr. Angela Lewis	Anxiety disorder.	Generalized Anxiety Disorder	Sertraline 50mg daily.

4. No of samples:5000, No of columns: 10, No of samples generated: 3604

```
2025-05-07 16:46:55,725 - INFO - Initiating generation for category: Tabular
2025-05-07 16:46:55,725 - INFO - Using TabularDataGenerator.
2025-05-07 16:46:55,725 - INFO - Initializing Azure Open AI Model for tabular data
2025-05-07 16:46:55,726 - INFO - Initializing the Azure OpenAI model.
2025-05-07 16:46:55,726 - INFO - Azure OpenAI configuration loaded successfully.
2025-05-07 16:46:55,727 - INFO - No environment configuration found.
2025-05-07 16:46:55,727 - INFO - ManagedIdentityCredential will use IMDS
2025-05-07 16:46:55,728 - INFO - Token retrieved successfully.
2025-05-07 16:46:55,893 - INFO - Azure OpenAI LLM model initialized successfully.
2025-05-07 16:46:55,895 - INFO - Tokenizer initialized for model: gpt-4o-mini
2025-05-07 16:46:55,919 - INFO - Initializing the Azure OpenAI model.
2025-05-07 16:46:55,923 - INFO - Azure OpenAI configuration loaded successfully.
2025-05-07 16:46:55,923 - INFO - Tokenizer initialized for model: gpt-4o-mini
2025-05-07 16:46:55,927 - INFO - Schema token count: 298
2025-05-07 16:46:55,933 - INFO - Prompt token count: 2293
2025-05-07 16:46:55,937 - INFO - Prompt Sample token count: 2293
2025-05-07 16:46:55,937 - INFO - Using adaptive batch size: 52
2025-05-07 16:46:55,938 - INFO - Processing batch 1 with 52 records
2025-05-07 16:46:55,944 - INFO - Prompt token count: 2293
```

```
2025-05-07 18:25:49,188 - INFO - Final DataFrame with 3604 records.
2025-05-07 18:25:49,191 - INFO - Total 3604 unique records generated.
2025-05-07 18:25:49,194 - INFO - Total 0 duplicate records skipped.
2025-05-07 18:25:49,196 - INFO - 0 batches failed.
2025-05-07 18:25:49,198 - INFO - Average records per batch: 37.15
2025-05-07 18:25:49,198 - INFO - Cumulative retry count: 0
2025-05-07 18:25:49,200 - INFO - Saving synthetic data as: tabular_data.csv
2025-05-07 18:25:49,240 - INFO - Synthetic tabular data saved to: D:\Users\ry_pipeline\Data\tabular_synthetic_data\tabular_20250507_182549.csv
```

Generate Data

Start Generation

Data generated successfully! 3604 records generated. Here's a Preview:

	Patient name	Patient age	Department visited	Name of the doctor	Medical history	Diagnosis	Prescribed treatment	Treatment summary
0	John Doe	45	Cardiology	Dr. Emily Carter	Hypertension, high cholesterol	Coronary artery disease	Statins, beta-blockers, lifestyle changes	Patient diagnosed with coronary artery disease.
1	Alice Smith	38	Dermatology	Dr. Michael Brown	Eczema, allergies	Atopic dermatitis	Topical corticosteroids, antihistamines	Patient presents with atopic dermatitis. Recommended treatment includes topical steroids and antihistamines.
2	Robert Johnson	54	Orthopedics	Dr. Sarah Wilson	Previous knee surgery	Osteoarthritis	Physical therapy, NSAIDs	Patient diagnosed with osteoarthritis in the right knee. Recommended treatment includes physical therapy and NSAIDs.
3	Jessica Lee	29	Gynecology	Dr. Karen Davis	Irregular periods	Polycystic ovary syndrome (PCOS)	Hormonal therapy, lifestyle modification	Patient diagnosed with PCOS. Prescribed hormonal therapy and lifestyle changes.
4	Michael Brown	62	Gastroenterology	Dr. John Smith	Chronic liver disease	Cirrhosis	Liver function monitoring, dietary changes	Patient diagnosed with cirrhosis. Recommended treatment includes liver function monitoring and dietary changes.
5	Linda Garcia	47	Endocrinology	Dr. Nancy Thomas	Diabetes	Type 2 diabetes	Metformin, dietary changes	Patient diagnosed with Type 2 diabetes. Prescribed Metformin and dietary changes.
6	David Martinez	50	Neurology	Dr. Lisa White	Migraines	Chronic migraines	Preventive medication, lifestyle changes	Patient diagnosed with chronic migraines. Recommended treatment includes preventive medication and lifestyle changes.
7	Sarah Wilson	34	Psychiatry	Dr. James Green	Anxiety disorder	Generalized anxiety disorder	Cognitive behavioral therapy, SSRIs	Patient diagnosed with generalized anxiety disorder. Recommended treatment includes CBT and SSRIs.
8	Daniel Thompson	41	Pulmonology	Dr. Patricia Hall	Asthma	Persistent asthma	Inhaled corticosteroids, bronchodilators	Patient diagnosed with persistent asthma. Recommended treatment includes inhaled corticosteroids and bronchodilators.
9	Karen Robinson	56	Oncology	Dr. Richard Young	Breast cancer survivor	Breast cancer recurrence	Chemotherapy, radiation therapy	Patient diagnosed with breast cancer recurrence. Recommended treatment includes chemotherapy and radiation therapy.